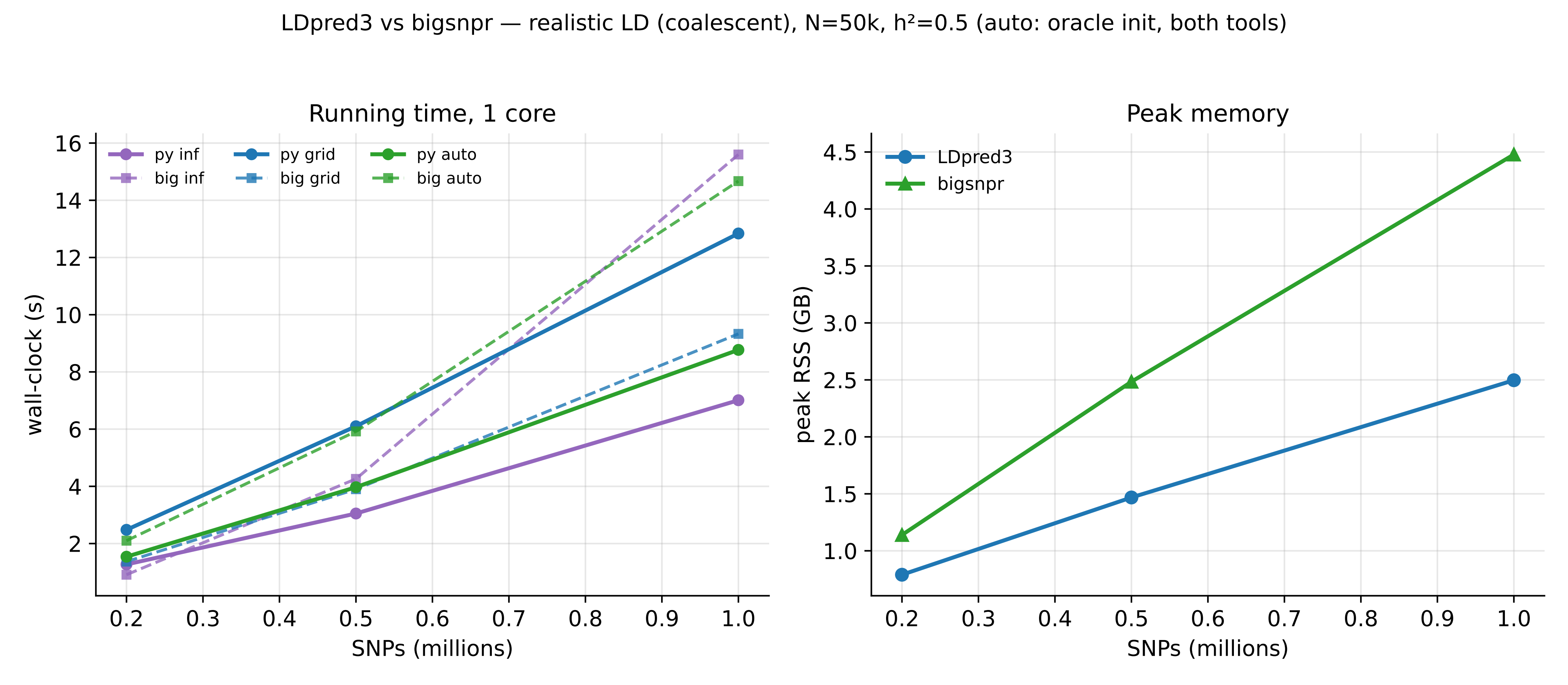
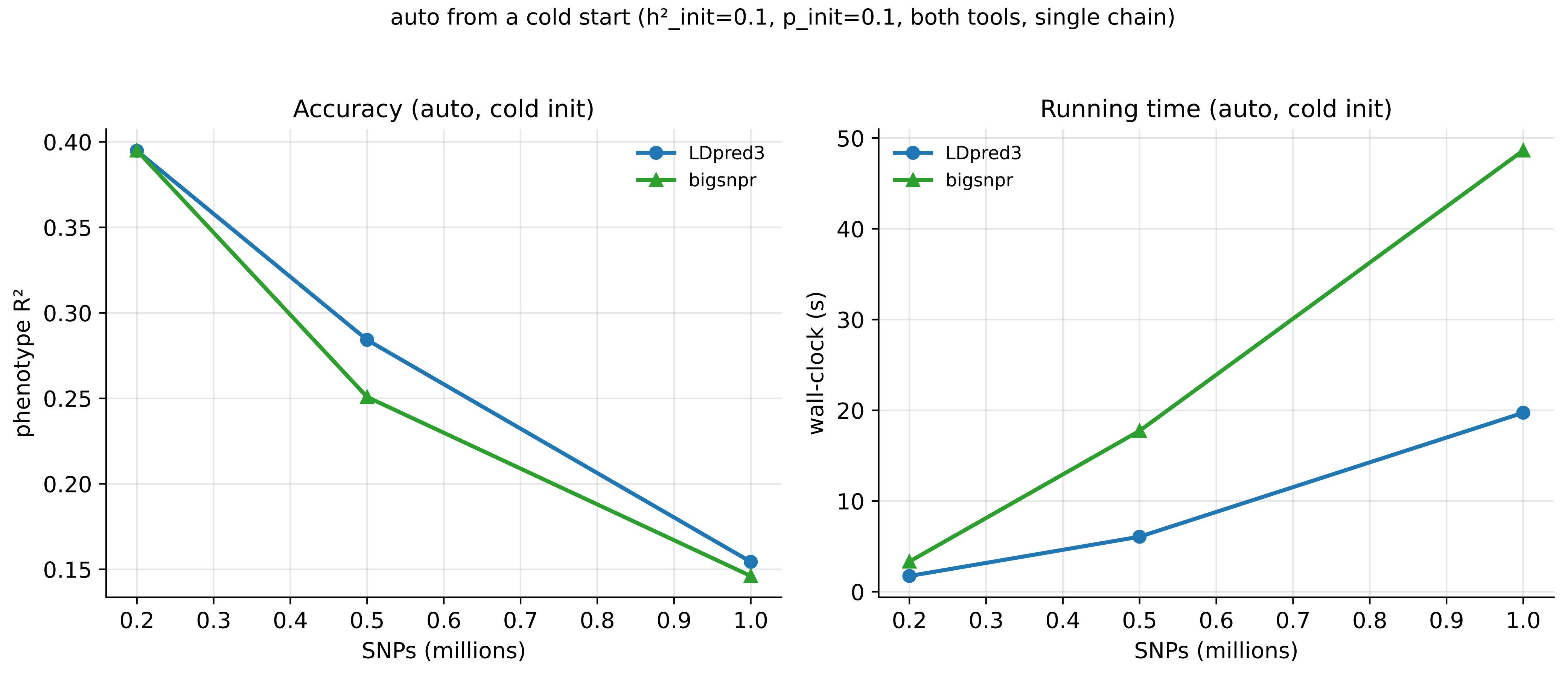
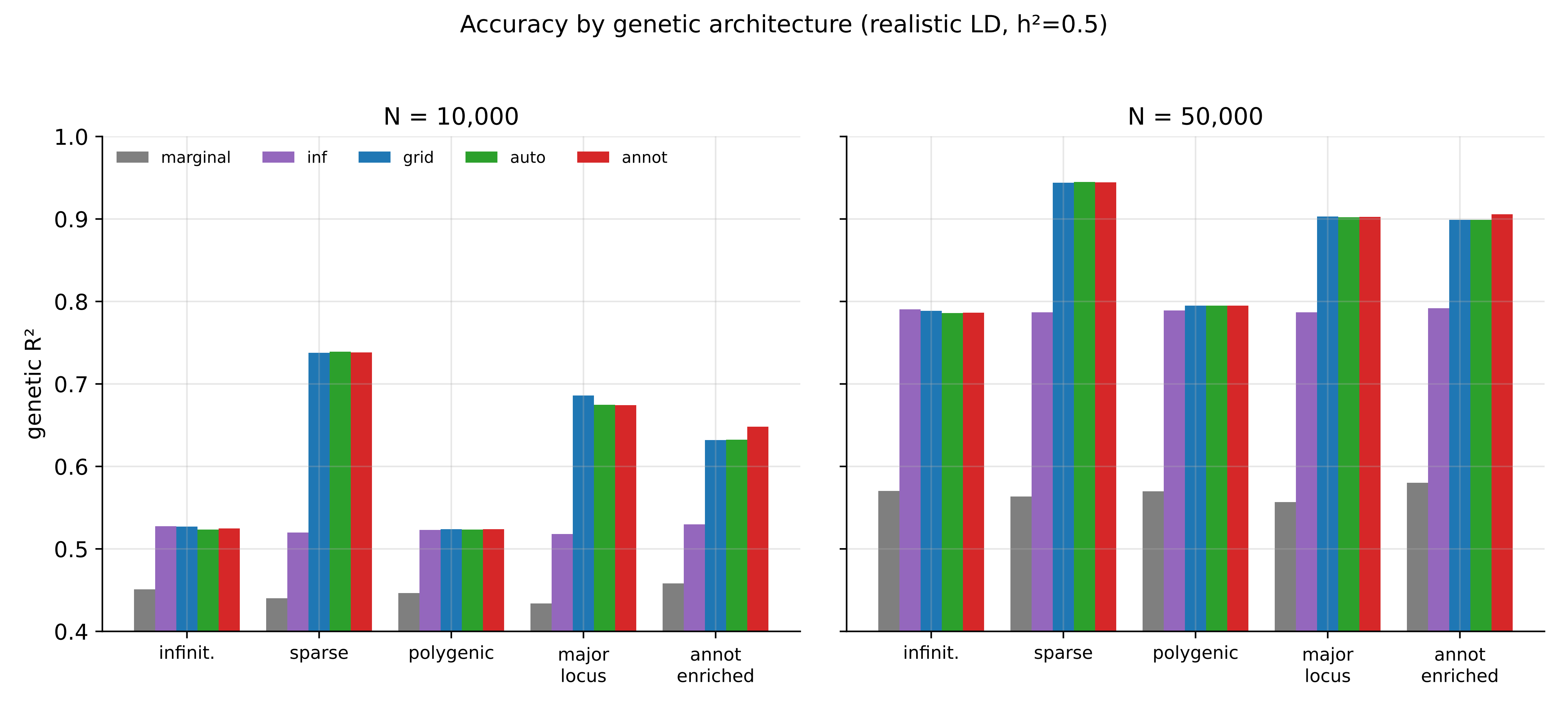
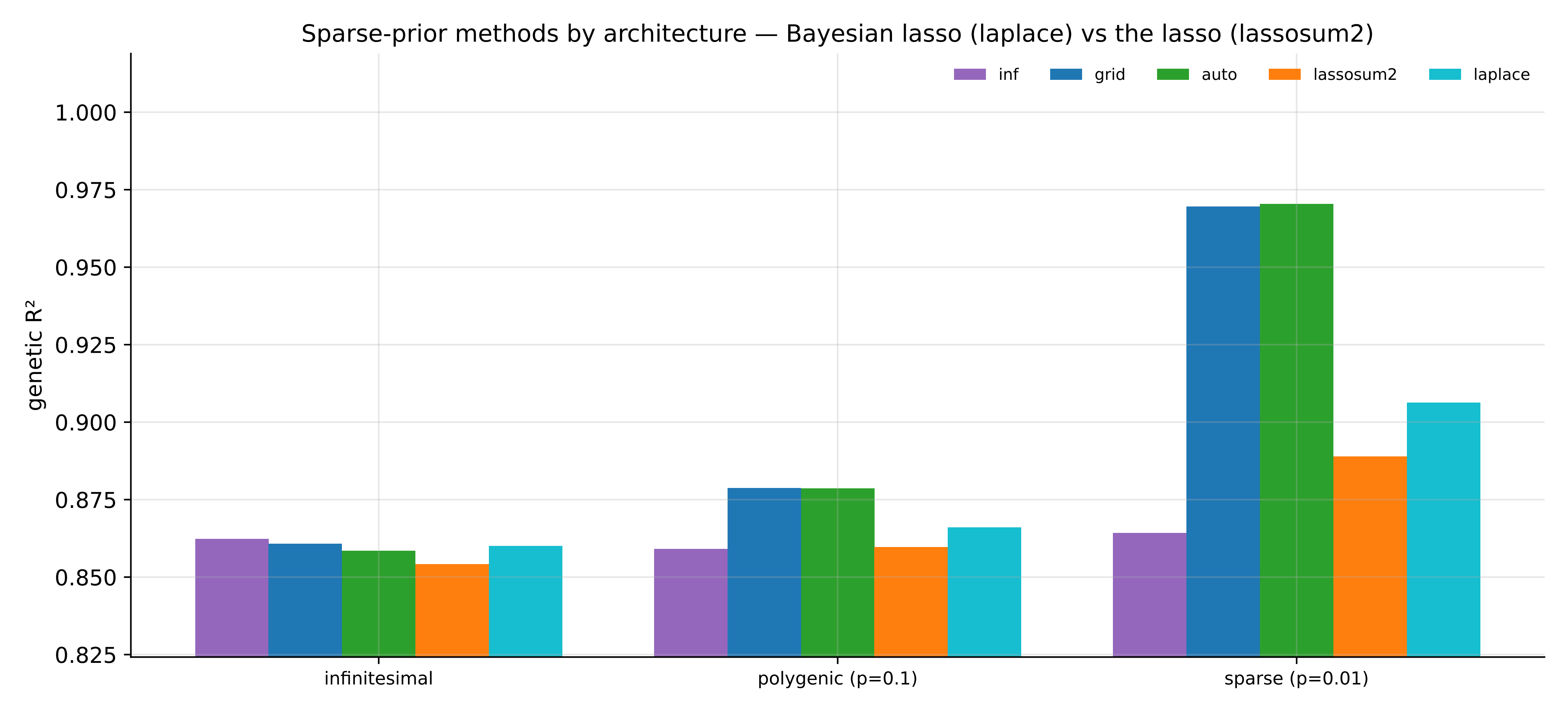
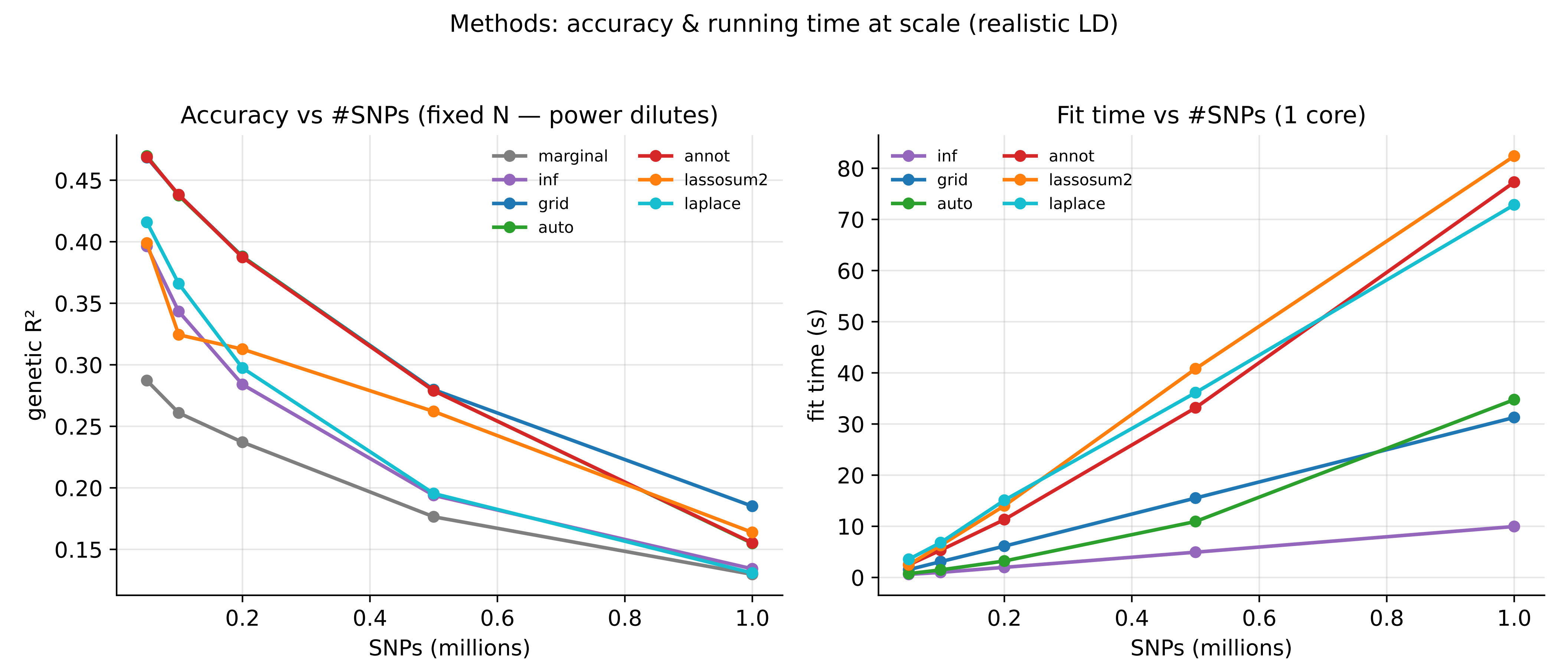

LDpred3 vs bigsnpr — realistic LD (coalescent), N=50k, h²=0.5 (auto: oracle init, both tools)
Running time, 1 core
wall-clock (s)
SNPs (millions)
py inf
big inf
py grid
big grid
py auto
big auto
Peak memory
peak RSS (GB)
SNPs (millions)
LDpred3
bigsnpr



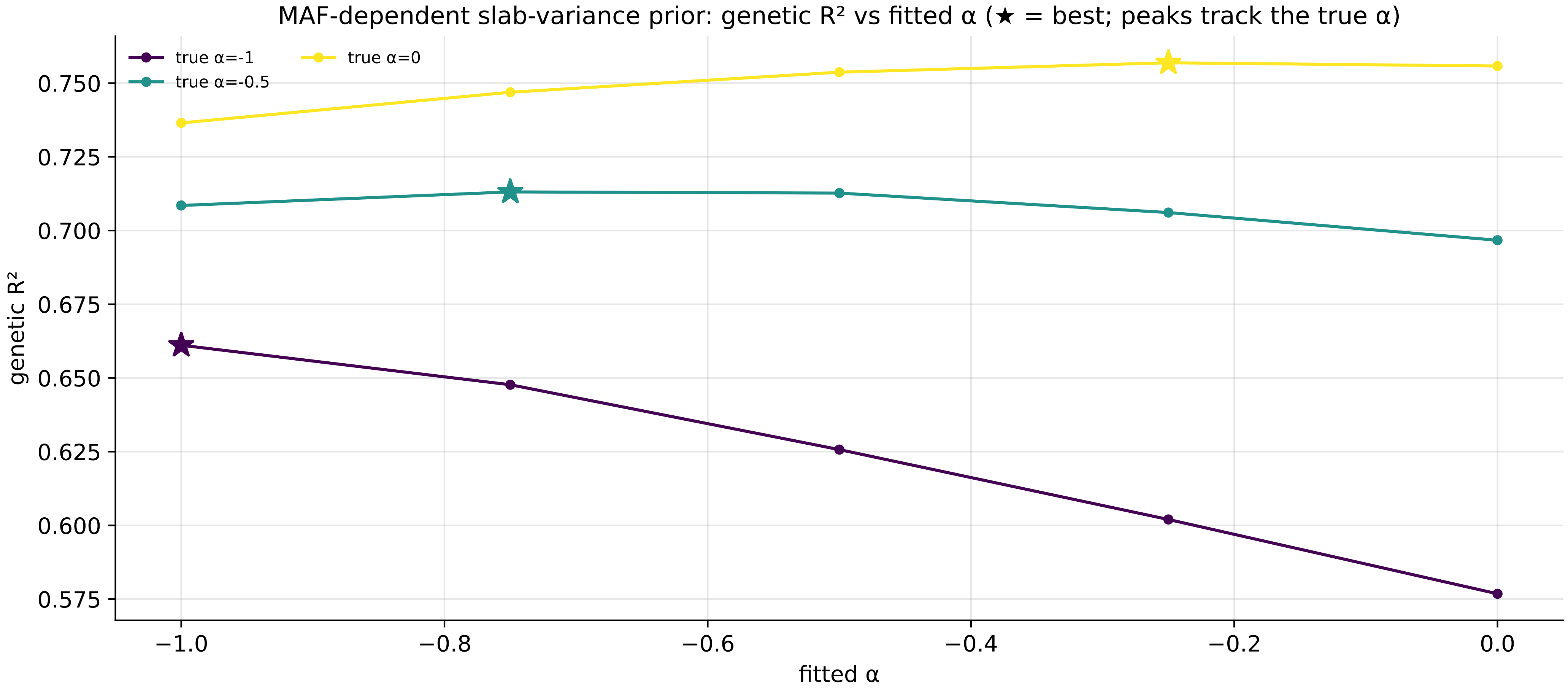
auto from a cold start (h²_init=0.1, p_init=0.1, both tools, single chain)
Accuracy (auto, cold init)
phenotype R²
0.40
0.35
0.30
0.25
0.20
0.15
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
SNPs (millions)
LDpred3
bigsnpr
Running time (auto, cold init)
wall-clock (s)
50
40
30
20
10
0
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
SNPs (millions)
LDpred3
bigsnpr


Accuracy by genetic architecture (realistic LD, h²=0.5)
N = 10,000
N = 50,000
genetic R²
1.0
0.9
0.8
0.7
0.6
0.5
0.4
marginal
inf
grid
auto
annot
infinit.
sparse
polygenic
major locus
annot enriched

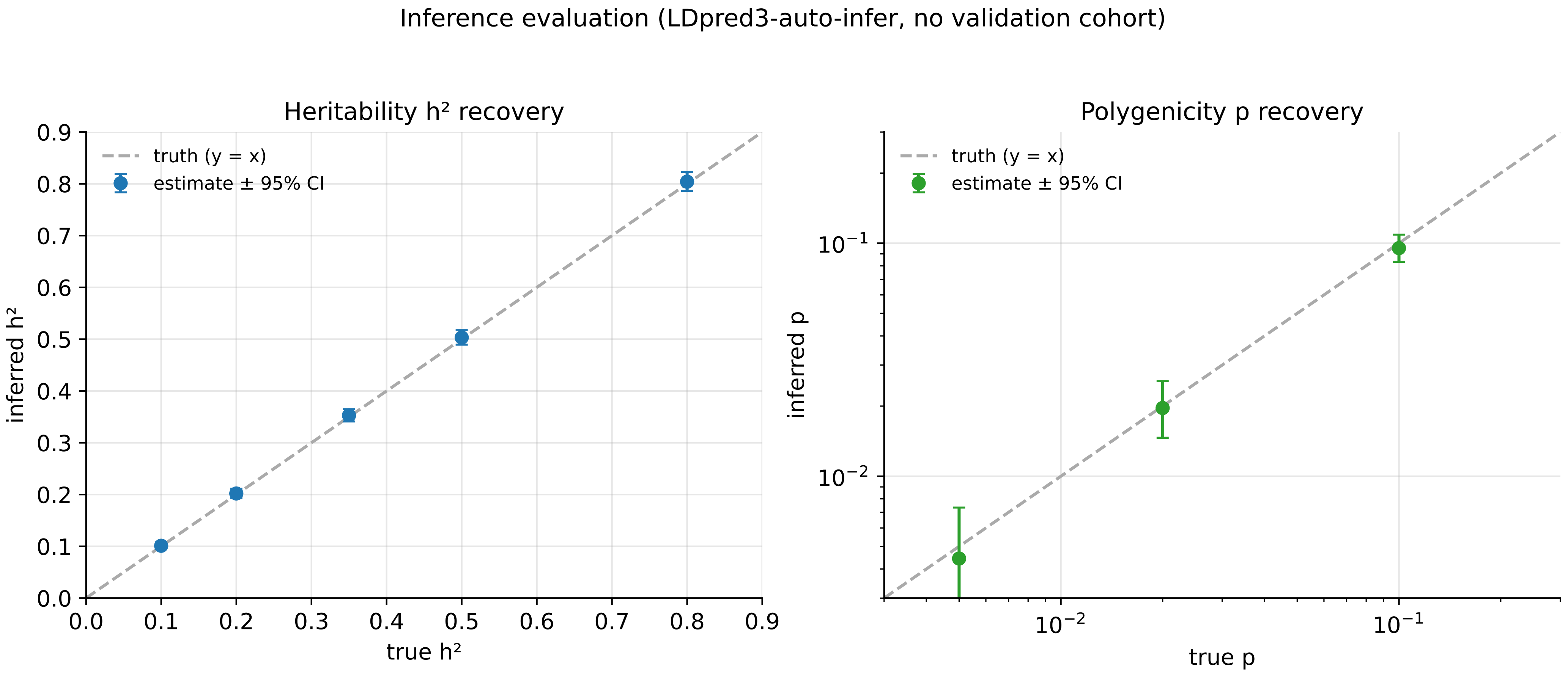

Sparse-prior methods by architecture — Bayesian lasso (laplace) vs the lasso (lassosum2)
inf
grid
auto
lassosum2
laplace
genetic R²
1.000
0.975
0.950
0.925
0.900
0.875
0.850
0.825
infinitesimal
polygenic (p=0.1)
sparse (p=0.01)



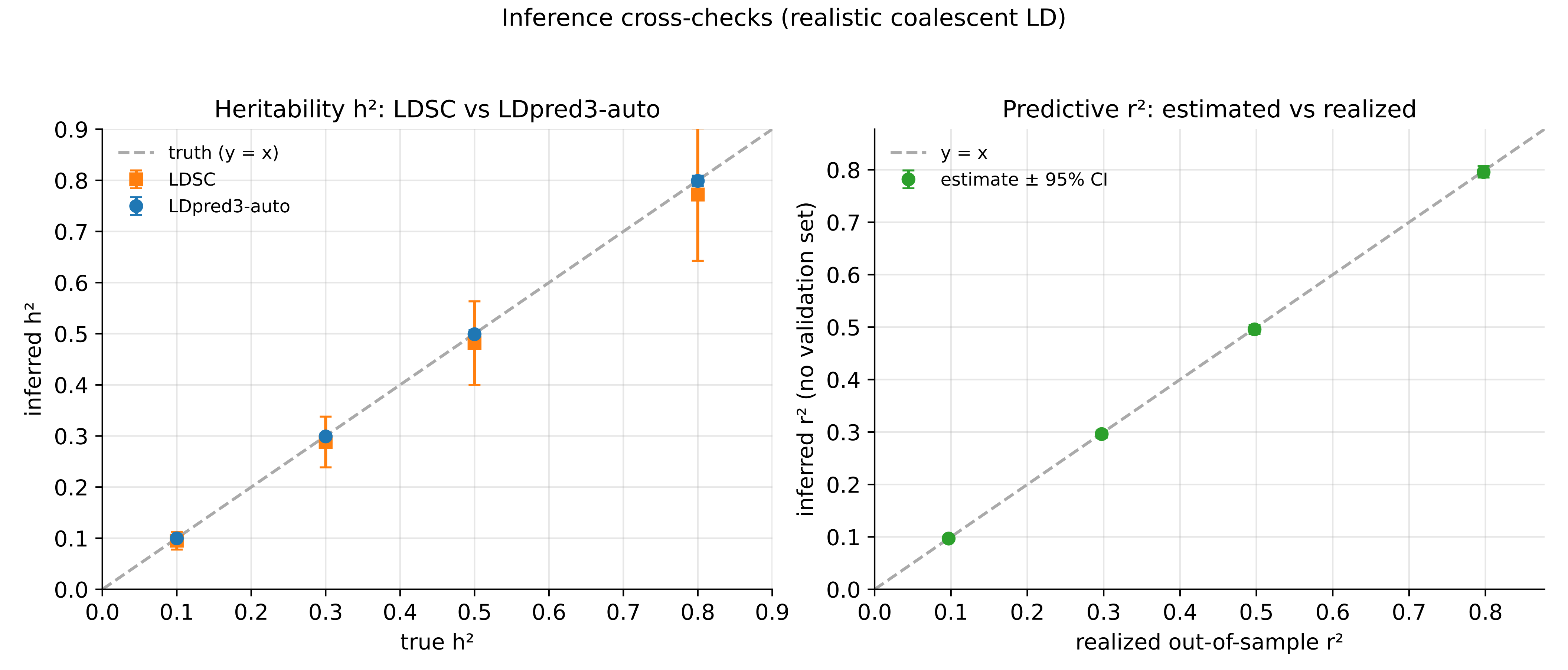
Methods: accuracy & running time at scale (realistic LD)
Accuracy vs #SNPs (fixed N — power dilutes)
genetic R²
SNPs (millions)
marginal
inf
grid
auto
annot
lassosum2
laplace
0.15
0.20
0.25
0.30
0.35
0.40
0.45
0.2
0.4
0.6
0.8
1.0
Fit time vs #SNPs (1 core)
fit time (s)
SNPs (millions)
inf
grid
auto
annot
lassosum2
laplace
0
10
20
30
40
50
60
70
80
0.2
0.4
0.6
0.8
1.0



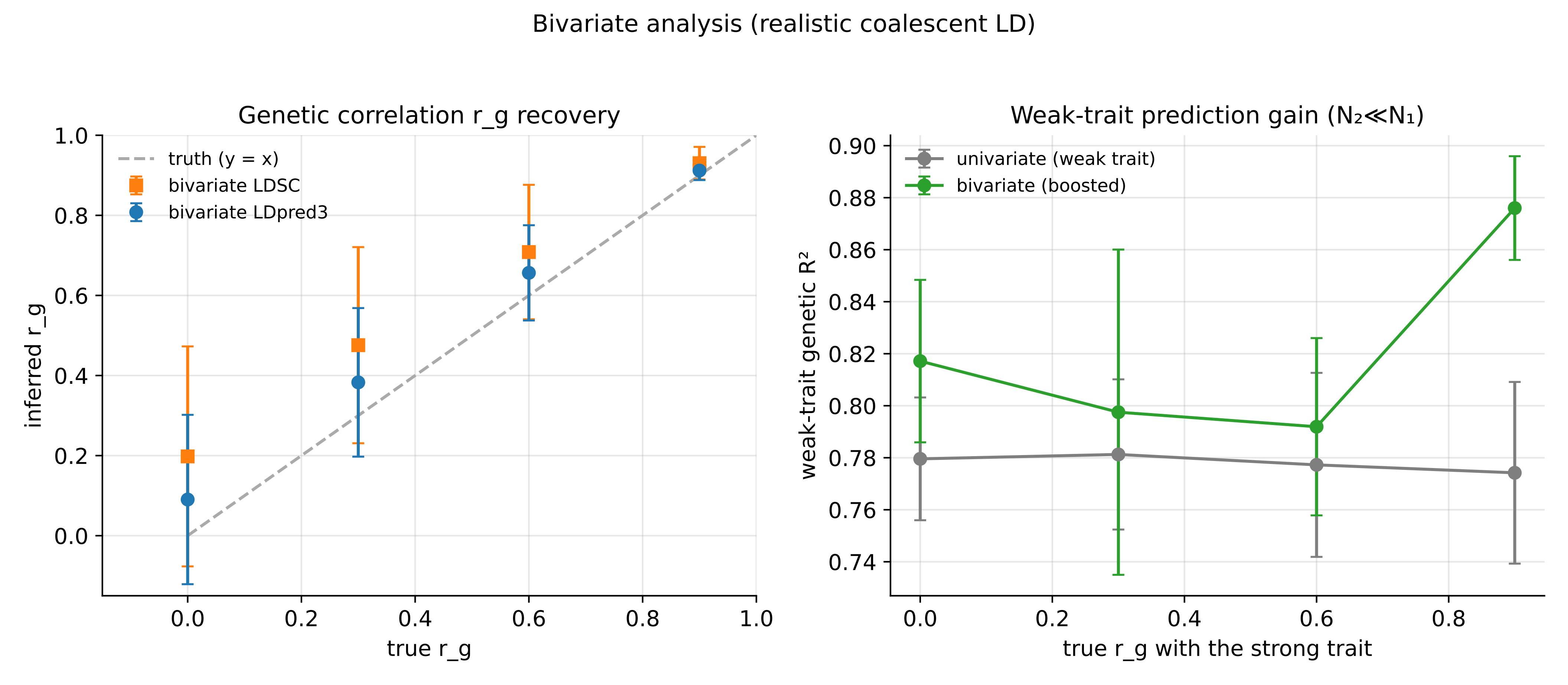
MAF-dependent slab-variance prior: genetic R² vs fitted α (★ = best; peaks track the true α)
true α=-1
true α=-0.5
true α=0
0.750
0.725
0.700
0.675
0.650
0.625
0.600
0.575
genetic R²
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
fitted α

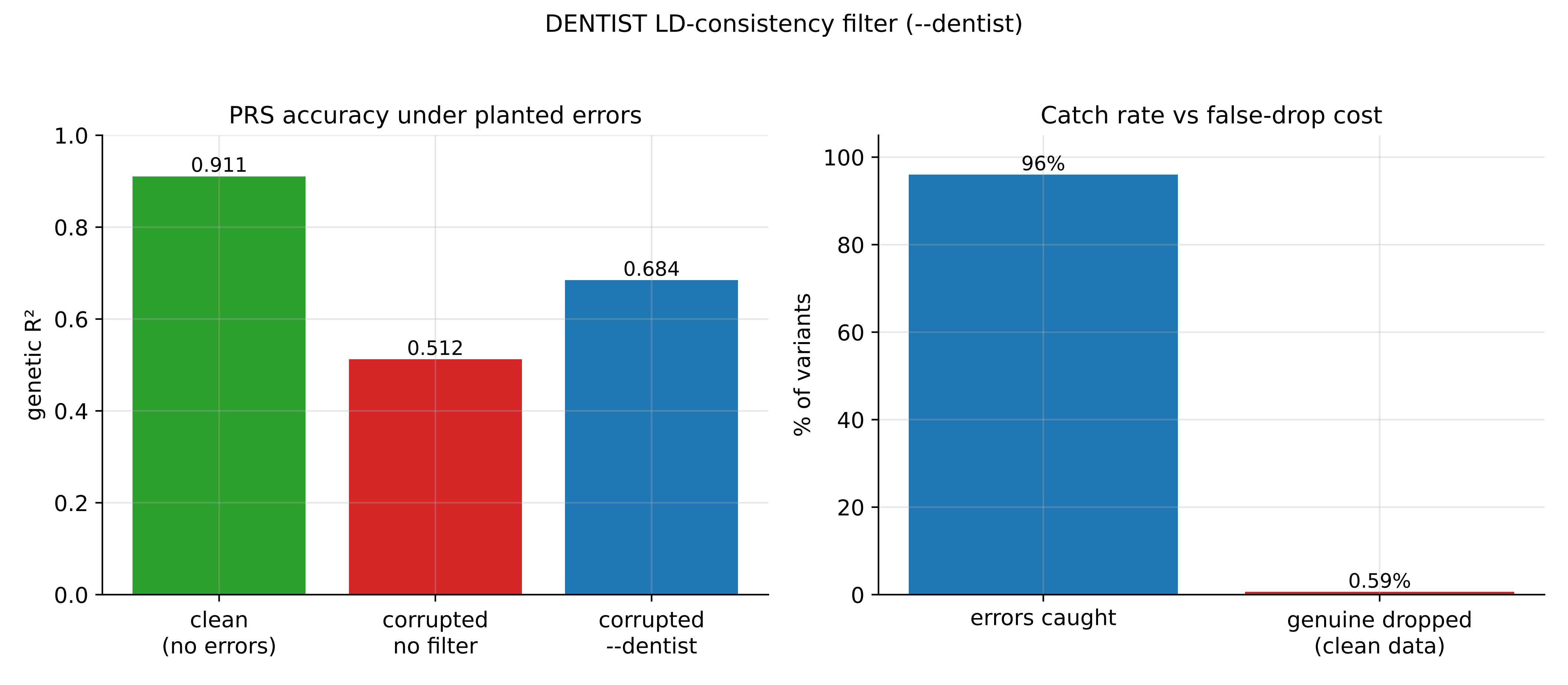


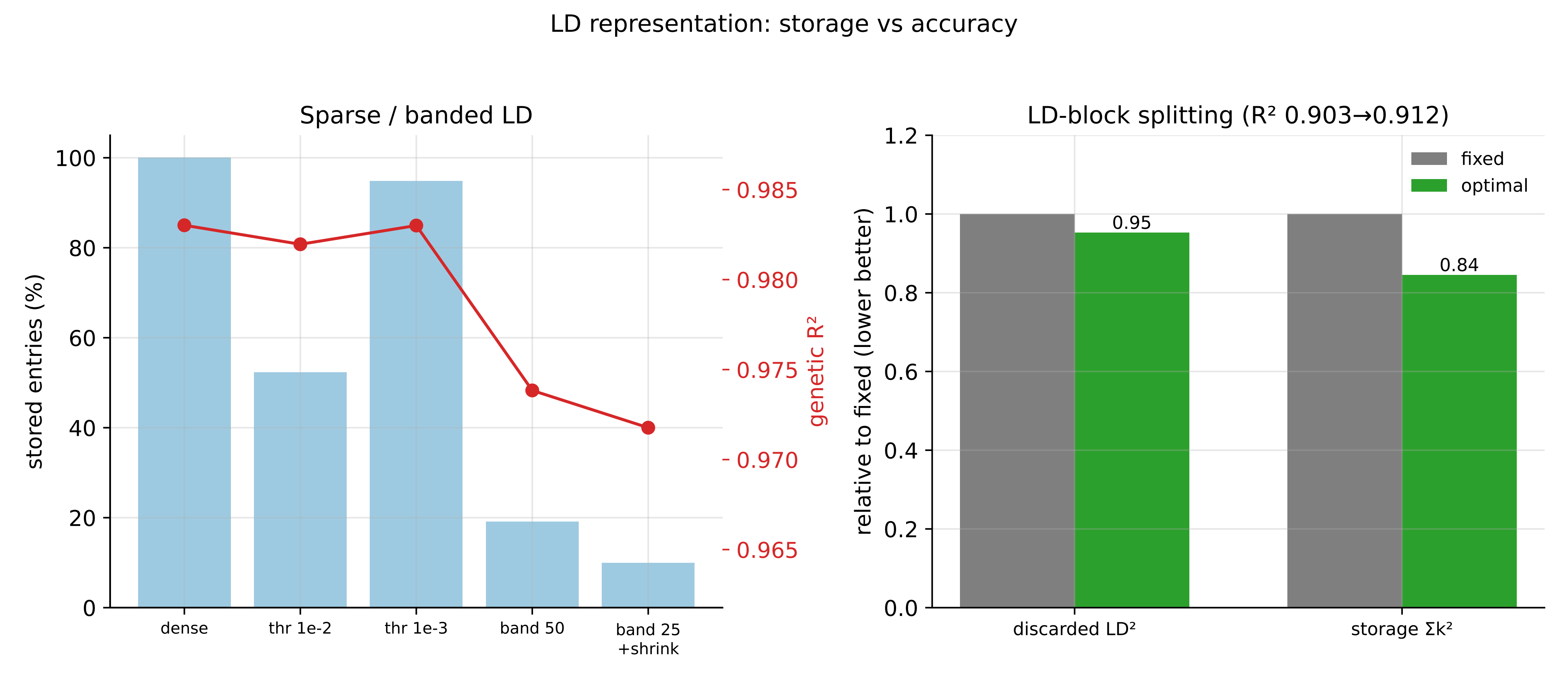
Inference evaluation (LDpred3-auto-infer, no validation cohort)
Heritability h² recovery
truth (y = x)
estimate ± 95% CI
inferred h²
true h²
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
Polygenicity p recovery
truth (y = x)
estimate ± 95% CI
inferred p
true p
10^−1
10^−2



Inference cross-checks (realistic coalescent LD)
Heritability h²: LDSC vs LDpred3-auto
truth (y = x)
LDSC
LDpred3-auto
inferred h²
true h²
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
Predictive r²: estimated vs realized
y = x
estimate ± 95% CI
inferred r² (no validation set)
realized out-of-sample r²

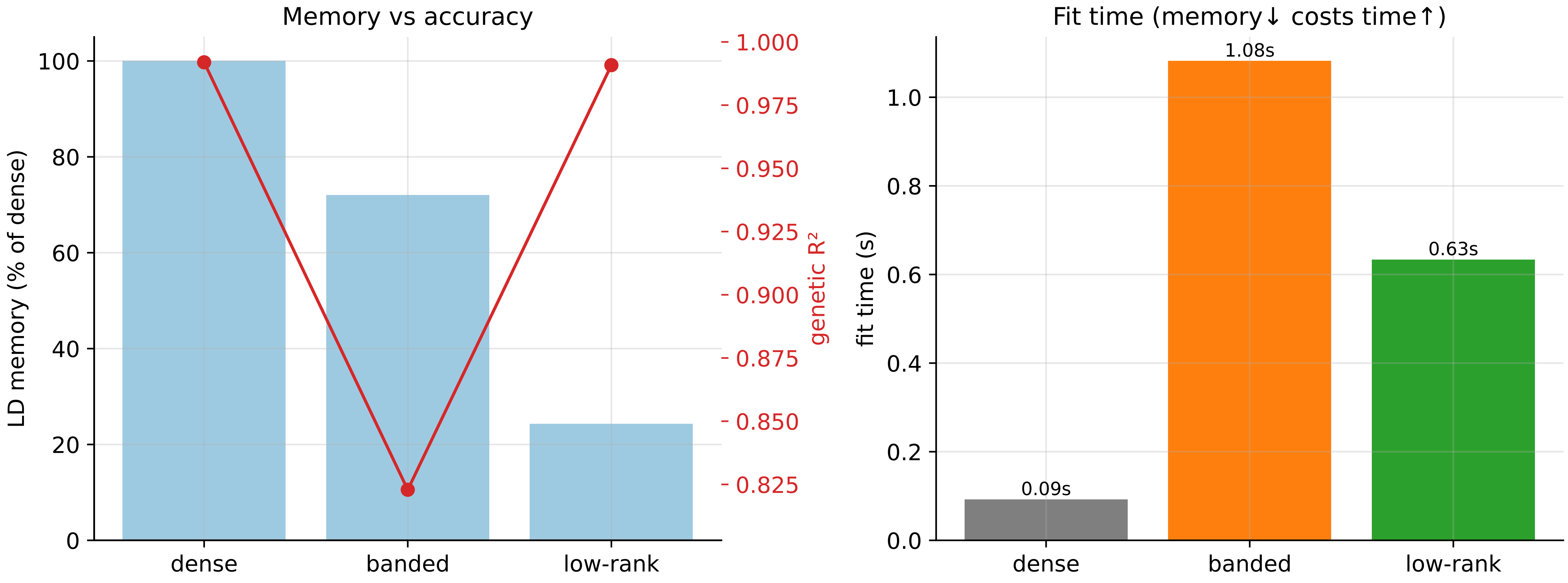

Bivariate analysis (realistic coalescent LD)
Genetic correlation r_g recovery
truth (y = x)
bivariate LDSC
bivariate LDpred3
inferred r_g
true r_g
Weak-trait prediction gain (N₂≪N₁)
univariate (weak trait)
bivariate (boosted)
weak-trait genetic R²
true r_g with the strong trait
1.0
0.8
0.6
0.4
0.2
0.0
0.90
0.88
0.86
0.84
0.82
0.80
0.78
0.76
0.74

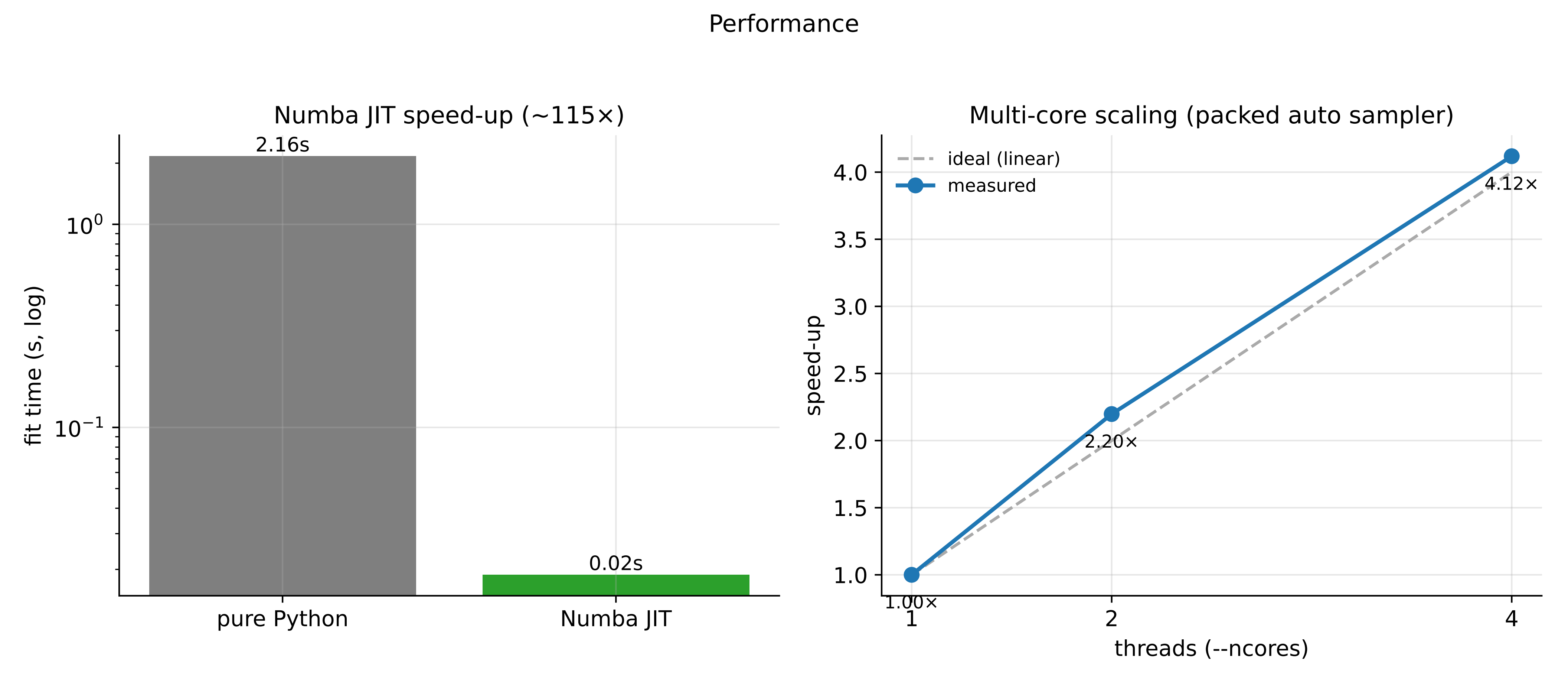

DENTIST LD-consistency filter (--dentist)
PRS accuracy under planted errors
genetic R²
1.0
0.8
0.6
0.4
0.2
0.0
0.911
0.512
0.684
clean
(no errors)
corrupted
no filter
corrupted
--dentist
Catch rate vs false-drop cost
% of variants
100
80
60
40
20
0
96%
0.59%
errors caught
genuine dropped
(clean data)



LD representation: storage vs accuracy
Sparse / banded LD
stored entries (%)
genetic R²
100
80
60
40
20
0
0.985
0.980
0.975
0.970
0.965
dense
thr 1e-2
thr 1e-3
band 50
band 25
+shrink
LD-block splitting (R² 0.903→0.912)
relative to fixed (lower better)
1.2
1.0
0.8
0.6
0.4
0.2
0.0
fixed
optimal
0.95
0.84
discarded LD²
storage Σk²

LD representations at scale — realistic coalescent LD (low-rank matches dense at ~1/4 memory; banding loses accuracy)





Performance
Numba JIT speed-up (~115×)
2.16s
0.02s
fit time (s, log)
10^0
10^{-1}
pure Python
Numba JIT
Multi-core scaling (packed auto sampler)
ideal (linear)
measured
4.12×
2.20×
1.00×
speed-up
4.0
3.5
3.0
2.5
2.0
1.5
1.0
1
2
4
threads (--ncores)